Project: Worked examples
Title: Tolerance stack-up (1D)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the worst-case, root-sum-square and modified RSS limits of a linear tolerance loop, with the predicted spread and capability. It adds a contribution Pareto and a Monte Carlo prediction against a stated requirement.

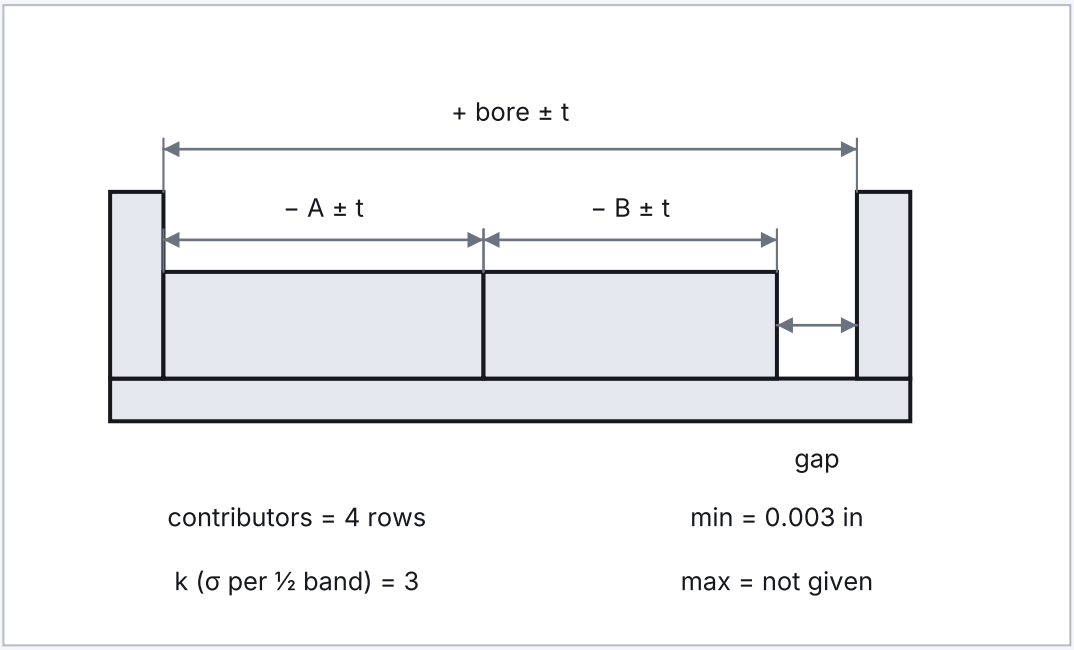


Figure 1 — A linear stack (representative; the contributors as listed): the bore adds, the parts subtract, the gap closes the loop. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Worst case, every contributor at its limit	$T_{wc} = \Sigma\lvert\text{half-bands}\rvert$
Root sum of squares, the statistical stack	$T_{rss} = \sqrt{\Sigma}\ \text{half}^2$
Modified RSS, a factor on the RSS (Bender-style)	$T_{mrss} = \text{factor} \times \text{RSS}$
Predicted standard deviation of the stack, from each contributor's distribution	$\sigma = \sqrt{\Sigma}\ \sigma_i^2$
Process capability, taken from the nearer limit	$C_{pk} = \min(USL - \mu, \mu - LSL)/(3\sigma)$
Monte Carlo check, every contributor sampled from a fixed seed and summed with its sign	$\text{stack} = \Sigma\ s_i\ x_i$

Basis: Technical product documentation and specification[1]; Dimensioning and tolerancing[2]; General tolerances (where a contributor's band is a general tolerance)[3].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
contributors	Contributors in the loop	4 row(s)		spec: Shigley 9e Ch. 1 example, part dimensions	any
notation	Drawing notation	ASME Y14.5		assumption: Inch drawing.	any
k	Standard deviations per half-band (normal contributors)	3		assumption: Each part at ±3σ within its band.	1 … 6
f_{rss}	Modified-RSS factor (Bender-style)	1.5		assumption: Bender factor 1.5.	1 … 2
cpk_{req}	Required C_{pk}	1.33		assumption: Usual capability target.	1 … 2
minimum	Minimum acceptable result (optional; blank = no minimum)	0.003	in	spec: Shigley 9e Ch. 1 example, required gap	-393.7 … 393.7 in
n_{mc}	Monte Carlo samples	10000		assumption: Default sample count.	1000 … 100000
seed	Monte Carlo seed	1		assumption: Fixed seed for reproducibility.	0 … 1000000000

TABLE ROW	VALUES
contributors[1]	name = a shoulder length; direction = adds; nominal = 1.75 in; upper = 0.003 in; lower = -0.003 in
contributors[2]	name = b sleeve; direction = subtracts; nominal = 0.75 in; upper = 0.001 in; lower = -0.001 in
contributors[3]	name = c washer; direction = subtracts; nominal = 0.12 in; upper = 0.005 in; lower = -0.005 in
contributors[4]	name = d sleeve; direction = subtracts; nominal = 0.875 in; upper = 0.001 in; lower = -0.001 in

3. Assumptions

A1. Every contributor acts along the one axis of the stack. (A linear stack: no angular or datum-shift effects beyond what the user enters as a contributor.)
A2. Contributors vary independently about their mid-limit and any stated mean shift, each with its stated distribution. (The worst case is a guarantee; every statistical figure is a prediction about a population.)
A3. A position or profile contributor contributes ± half its tolerance zone, plus half the stated MMC bonus. (Diametral zones are bilateral about the true position.)
A4. notation (Drawing notation) is taken as ASME Y14.5. (Inch drawing.)
A5. k (Standard deviations per half-band (normal contributors)) is taken as 3. (Each part at ±3σ within its band.)
A6. f_{rss} (Modified-RSS factor (Bender-style)) is taken as 1.5. (Bender factor 1.5.)
A7. cpk_{req} (Required C_{pk}) is taken as 1.33. (Usual capability target.)
A8. n_{mc} (Monte Carlo samples) is taken as 10000. (Default sample count.)

A9. seed (Monte Carlo seed) is taken as 1. (Fixed seed for reproducibility.)

4. Working

[6] c1 = + 44.45 +0.0762/-0.0762
= mid 44.45 mm, ± 0.0762 mm, σ 0.0254 mm (normal)
= 44.45 mm
a shoulder length

[7] c2 = − 19.05 +0.0254/-0.0254
= mid 19.05 mm, ± 0.0254 mm, σ 0.00846667 mm (normal)
= -19.05 mm
b sleeve

[8] c3 = − 3.048 +0.127/-0.127
= mid 3.048 mm, ± 0.127 mm, σ 0.0423333 mm (normal)
= -3.048 mm
c washer

[9] c4 = − 22.225 +0.0254/-0.0254
= mid 22.225 mm, ± 0.0254 mm, σ 0.00846667 mm (normal)
= -22.225 mm
d sleeve

[10] x_{nom} = Σ (sign × nominal)
= 44.45 − 19.05 − 3.048 − 22.225
= 0.127 mm
Nominal result

[11] x_{mean} = Σ (sign × mid-limit)
= 44.45 − 19.05 − 3.048 − 22.225
= 0.127 mm
Mean result

[12] t_{wc} = Σ half-bands
= 0.0762 + 0.0254 + 0.127 + 0.0254
= 0.254 mm
Worst-case tolerance (±)

[13] $x_{wc,min}$ = x_{mean} - t_{wc}
= 0.127 mm − 0.254 mm
= -0.127 mm
Worst-case minimum

[14] $x_{wc,max}$ = x_{mean} + t_{wc}
= 0.127 mm + 0.254 mm
= 0.381 mm
Worst-case maximum

[15] t_{rss} = √Σ half²
= √(0.0762² + 0.0254² + 0.127² + 0.0254²)
= 0.1524 mm
Root-sum-square tolerance (±)

[16] $x_{rss,min}$ = x_{mean} - t_{rss}
= 0.127 mm − 0.1524 mm
= -0.0254 mm
RSS minimum

[17] $x_{rss,max}$ = x_{mean} + t_{rss}
= 0.127 mm + 0.1524 mm
= 0.2794 mm
RSS maximum

[18] t_{mrss} = f_{rss} * t_{rss}
= 1.5 × 0.1524 mm
= 0.2286 mm
Modified RSS tolerance (±)

[19] $\sigma = \sqrt{\Sigma\ \sigma_i^2}$
$= \sqrt{(0.0254^2 + 0.00846667^2 + 0.0423333^2 + 0.00846667^2)}$
$= 0.0508$ mm
Predicted standard deviation of the result
[20] $m_{min} = x_{wc,min}$ - minimum
$= -0.127 - 0.0762$
$= -0.2032$ mm
Worst-case margin to the minimum
[21] $cpk = \min(x_{mean} - LSL, USL - x_{mean}) / (3\ \sigma)$
$= \min(0.0508, \infty) / (3 \times 0.0508)$
$= 0.33333$
Process capability index C_{pk}
[22] $dpmo = 10^6 \times P(outside)$
$= 10^6 \times \Phi$-tails at $\sigma = 0.0508$
$= 158660$
Predicted defects per million
[23] $yield = 1 - dpmo / 10^6$
$= 1 - 158660 / 10^6$
$= 84.134$ %
Predicted yield
[24] $u_{cpk} = cpk_{target} / cpk$
$= 1.33 / 0.33333$
$= 3.99$
Capability utilisation
[25] pareto = largest contributor (statistical)
= a shoulder length 25 %; b sleeve 2.78 %; c washer 69.4 %; d sleeve 2.78 %
= "c washer"
Contribution Pareto
[28] mc_{mean} = mean of 10000 samples (seed 1)
= Monte Carlo
$= 0.12705$ mm
Monte Carlo mean
[29] mc_{sigma} = sample standard deviation
= Monte Carlo
$= 0.050593$ mm
Monte Carlo σ
[30] mc_{p05} = 5 % quantile
= Monte Carlo
$= 0.044003$ mm
Monte Carlo 5 %
[31] mc_{p95} = 95 % quantile
= Monte Carlo
$= 0.21023$ mm
Monte Carlo 95 %
[32] mc_{yield} = samples inside the requirement
= 8461 of 10000
= 84.61 %
Monte Carlo yield

5. Results

SYMBOL	VALUE	DESCRIPTION
x_{nom}	0.1270 mm	Nominal result
x_{mean}	0.1270 mm	Mean result

SYMBOL	VALUE	DESCRIPTION
t_{wc}	0.2540 mm	Worst-case tolerance (±)
t_{rss}	0.1524 mm	RSS tolerance (±)
σ	0.05080 mm	Predicted σ
$x_{wc,min}$	-0.1270 mm	Worst-case minimum
$x_{wc,max}$	0.3810 mm	Worst-case maximum
t_{mrss}	0.2286 mm	Modified RSS tolerance (±)
m_{min}	-0.2032 mm	Worst-case margin to the minimum
cpk	0.33333	C_{pk}
dpmo	158660	DPMO (predicted)
yield	84.13 %	Yield (predicted)
mc_{mean}	0.1270 mm	Monte Carlo mean
mc_{sigma}	0.05059 mm	Monte Carlo σ
mc_{p05}	0.04400 mm	Monte Carlo 5 % quantile
mc_{p95}	0.2102 mm	Monte Carlo 95 % quantile
mc_{yield}	84.61 %	Monte Carlo yield
u_{cpk}	3.99	Capability utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Worst-case range against the requirement	-0.127 … 0.381 mm	0.0762 … open mm	—	
FAIL	C_{pk} against the requirement	1.33	0.33333	3.99	

7. Applicability

None.

8. Verification

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Template mech.tolerance-stack@2.3.0 Content hash 1b96b6f2b2031243 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Technical Product Documentation and Specification. BS 8888. BSI.
2. American Society of Mechanical Engineers. Dimensioning and Tolerancing. ASME Y14.5. ASME.
3. International Organization for Standardization. General Tolerances—Part 1: Tolerances for Linear and Angular Dimensions Without Individual Tolerance Indications. ISO 2768-1. ISO.